Course: SWE 4404 – Software Project Lab II

WANDERVILLE

Design of a Software Presentation



TEAM MEMBERS

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OVERVIEW

Wanderville is a **cozy 2D isometric multiplayer farming & social sim set** on a stylized version of campus — built for students who want a **relaxing, community-driven space.**

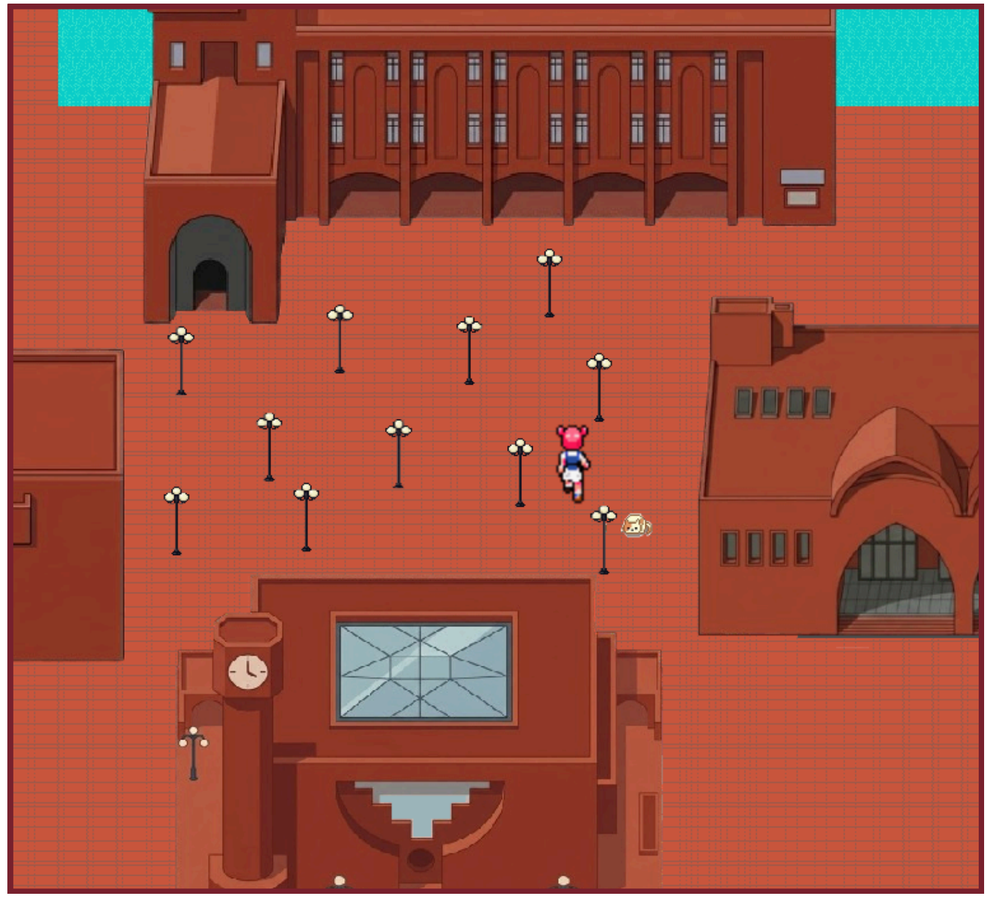
Our SRS :

- **126 functional requirements** across **20 feature areas**
- **Format:** FR ID → User Story → Acceptance Criteria → Priority → Status

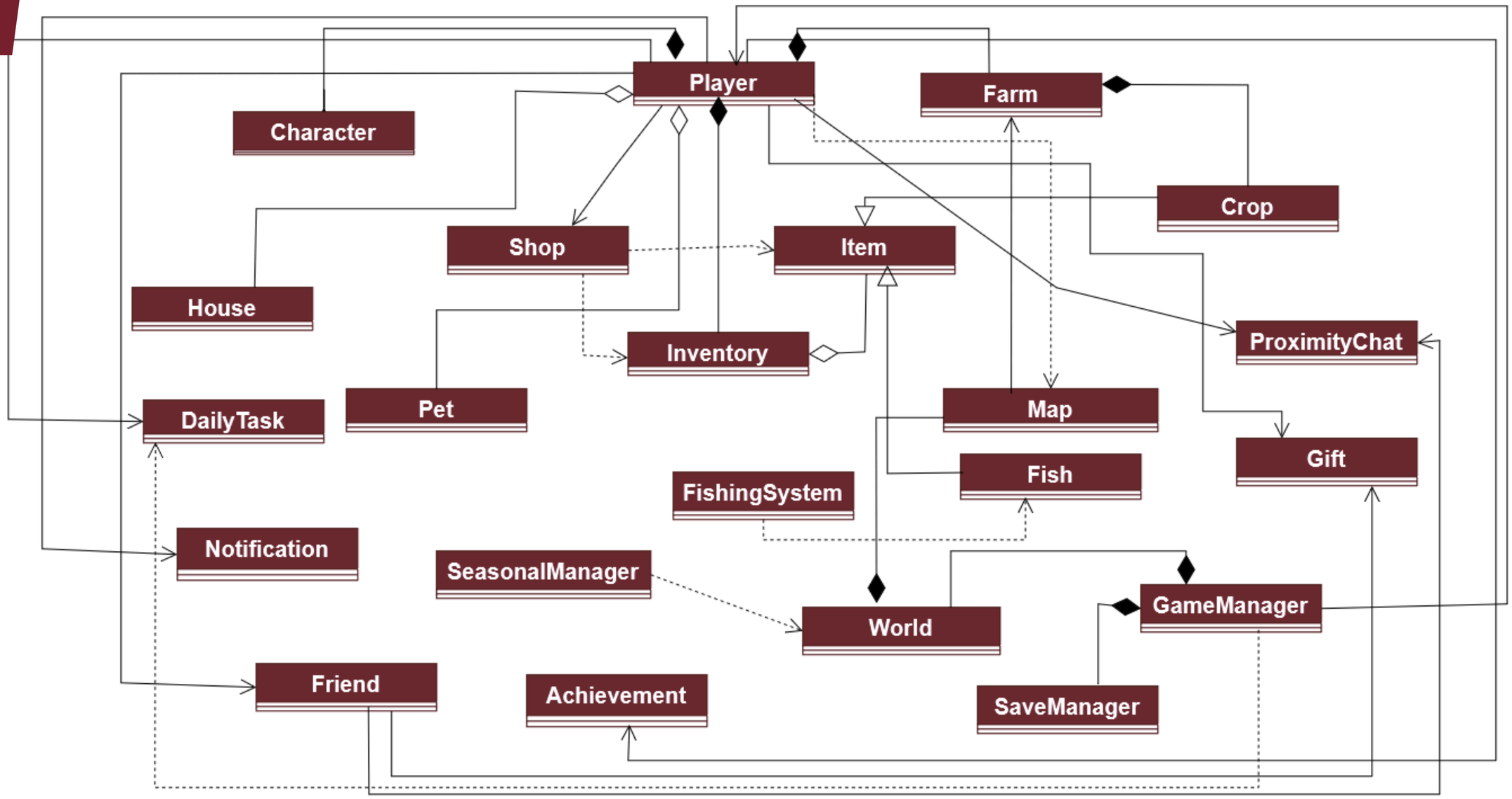


CLASS DIAGRAM

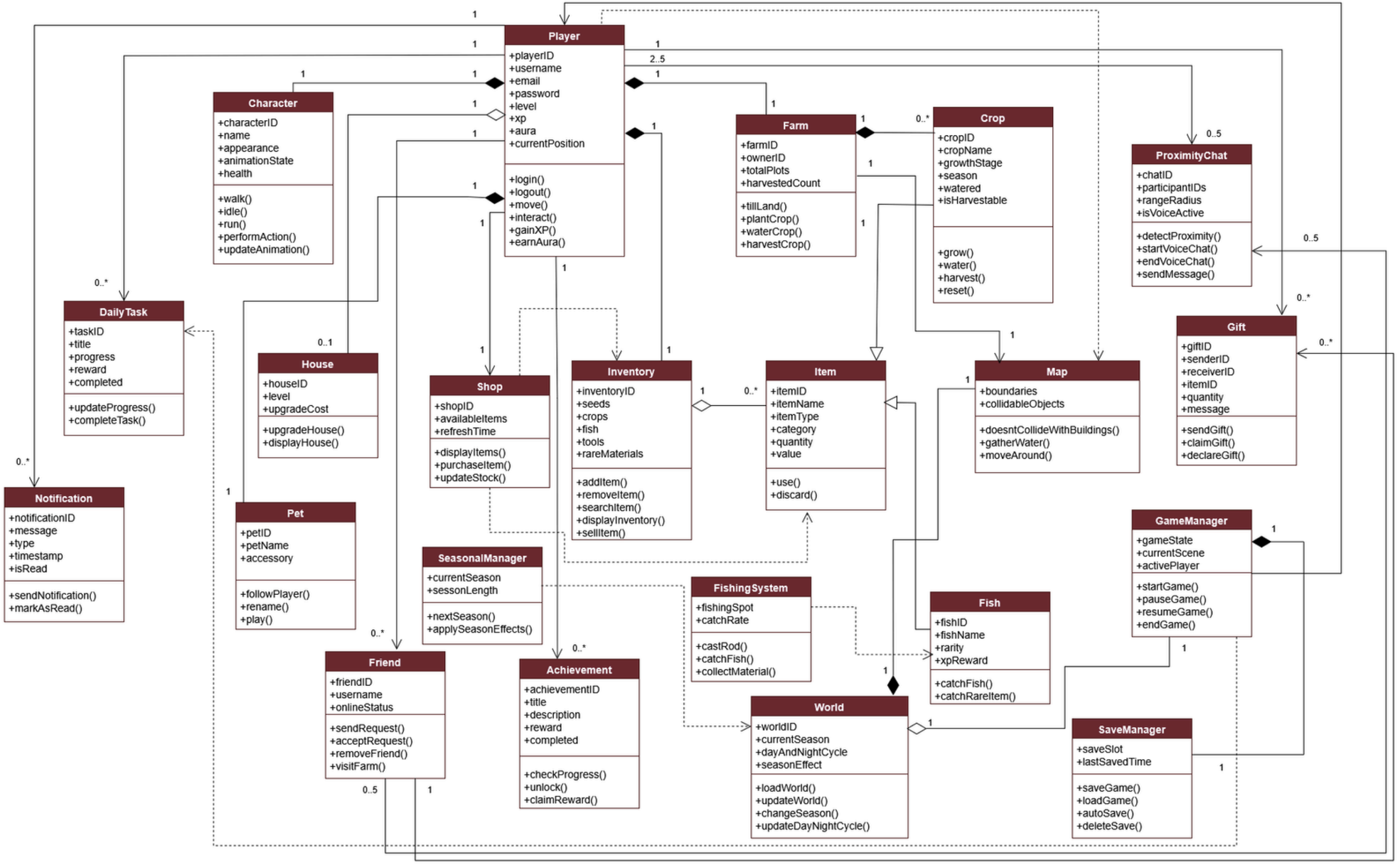
1	Static structure of the game - classes, attributes, methods, relationships
2	Derived from grouping related SRS requirements (e.g. farming FRs → Farm + Crop classes)
3	Player is the central class — connects to Inventory, Farm, Pet, Achievement
4	GameManager coordinates overall gameplay state



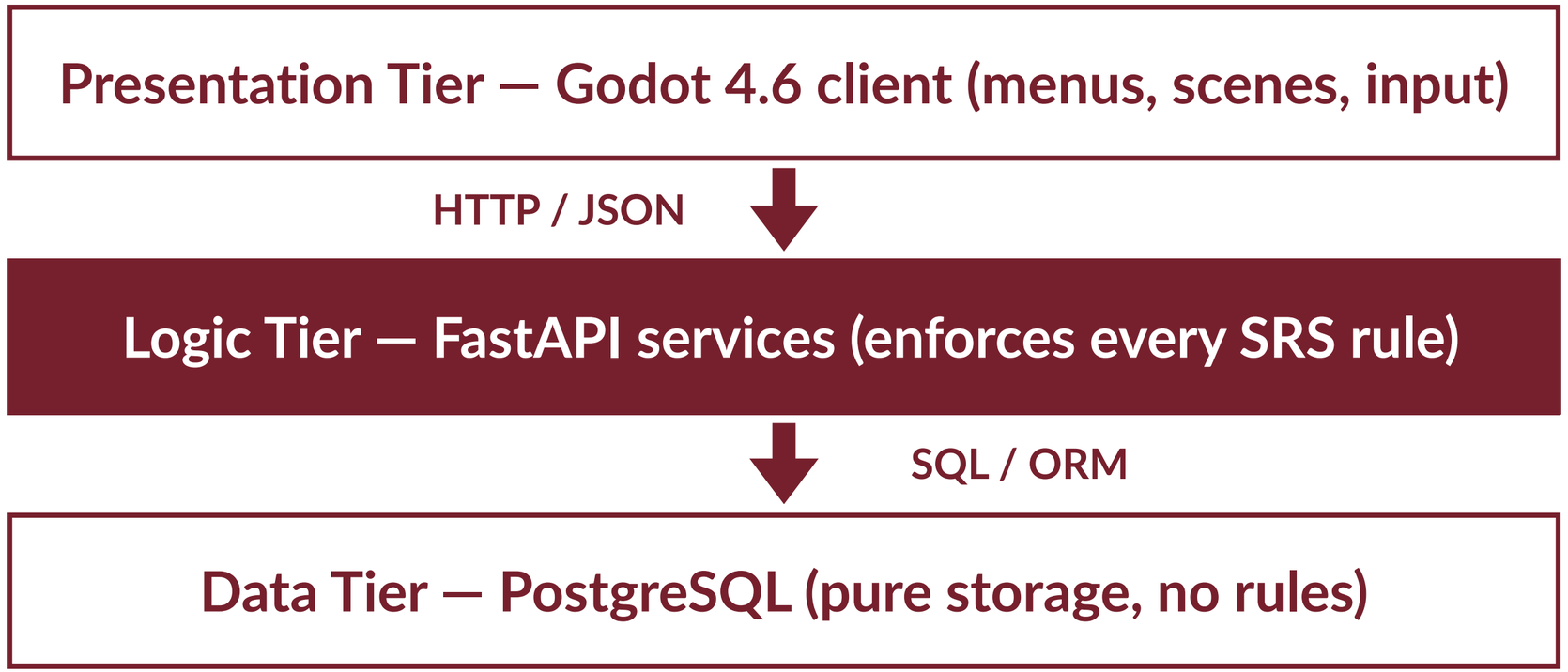
Class diagram



Class diagram (Detailed)



Why Three-Tier Architecture



Why? client never touches the database directly — every acceptance criterion is enforced in one place, the service layer.

Three-Tier Architecture

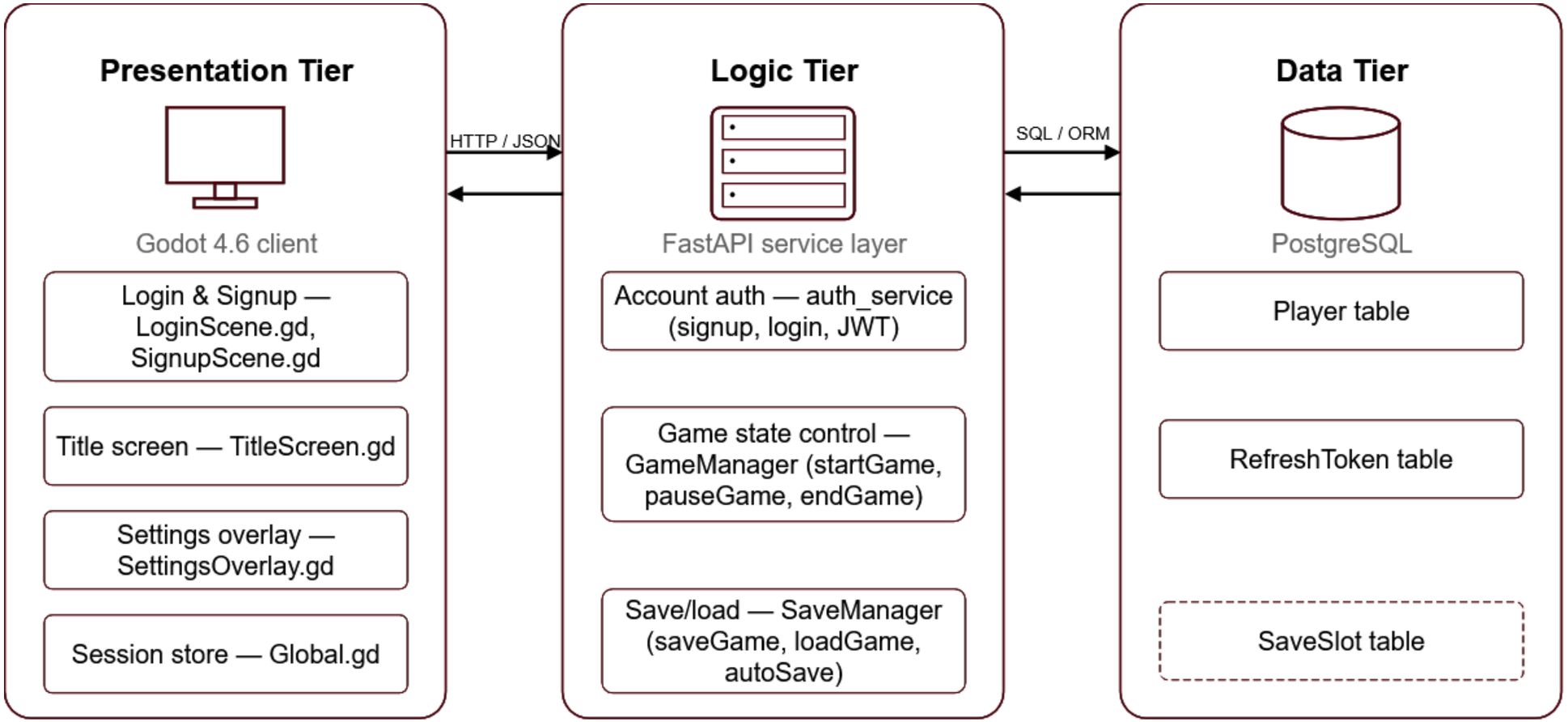
Domain-Based Architecture Split

- Project split into **5 domains**, each covering a related group of SRS requirements
- One **three-tier diagram per domain**, instead of one large diagram
- Keeps each diagram **readable** and shows **real implementation progress**
- Solid boxes = **built** · Dashed/grey boxes = designed in SRS, **not yet built**

Domains

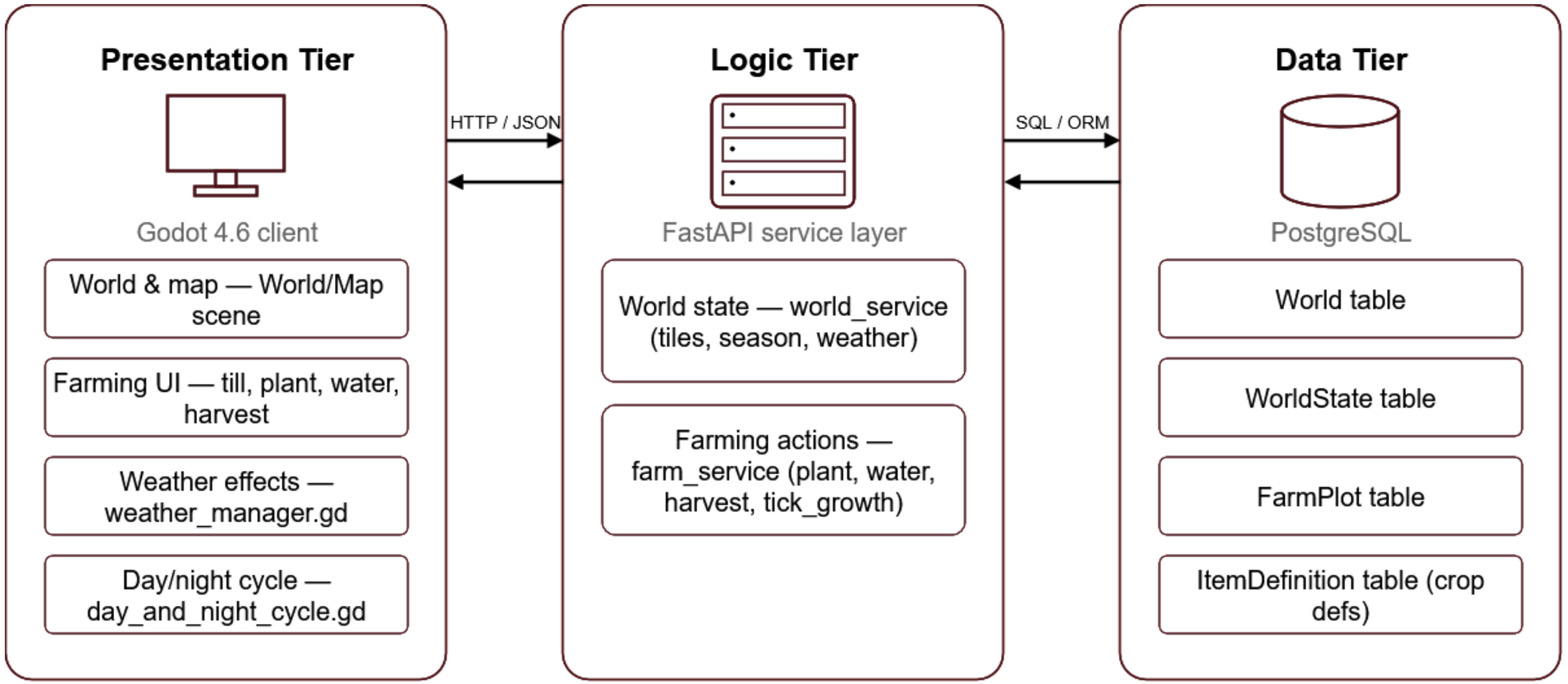
- **Authentication** & **Core** Session
- **World, Farming** & **Environment**
- **Inventory, Economy** & **Shop**
- **Fishing, Pets** & **Social**
- **Achievements, Daily Tasks** & **Notifications**

Architecture: Domain 1
Authentication & Core Session



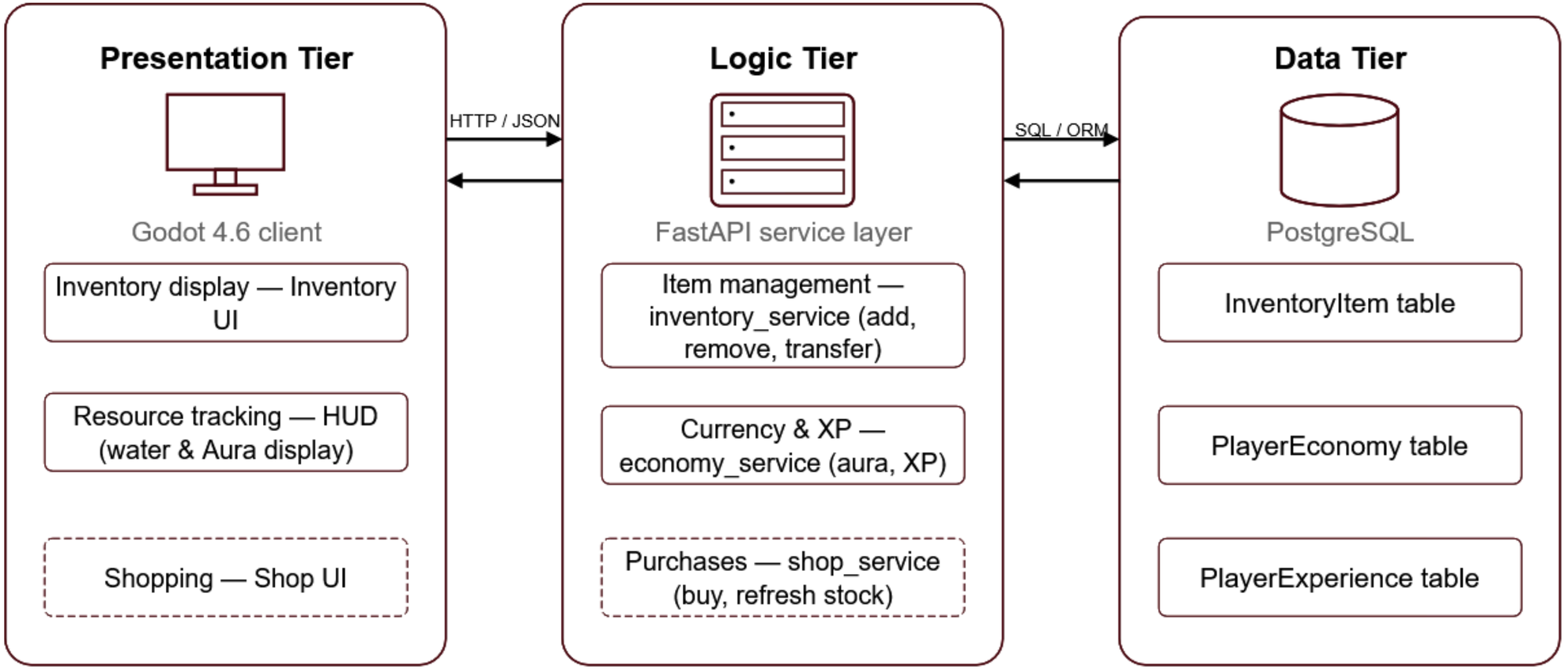
Handles account creation, login, session persistence, and save/load, backed by the Player and RefreshToken tables.

Architecture: Domain 2
World, Farming & Environment



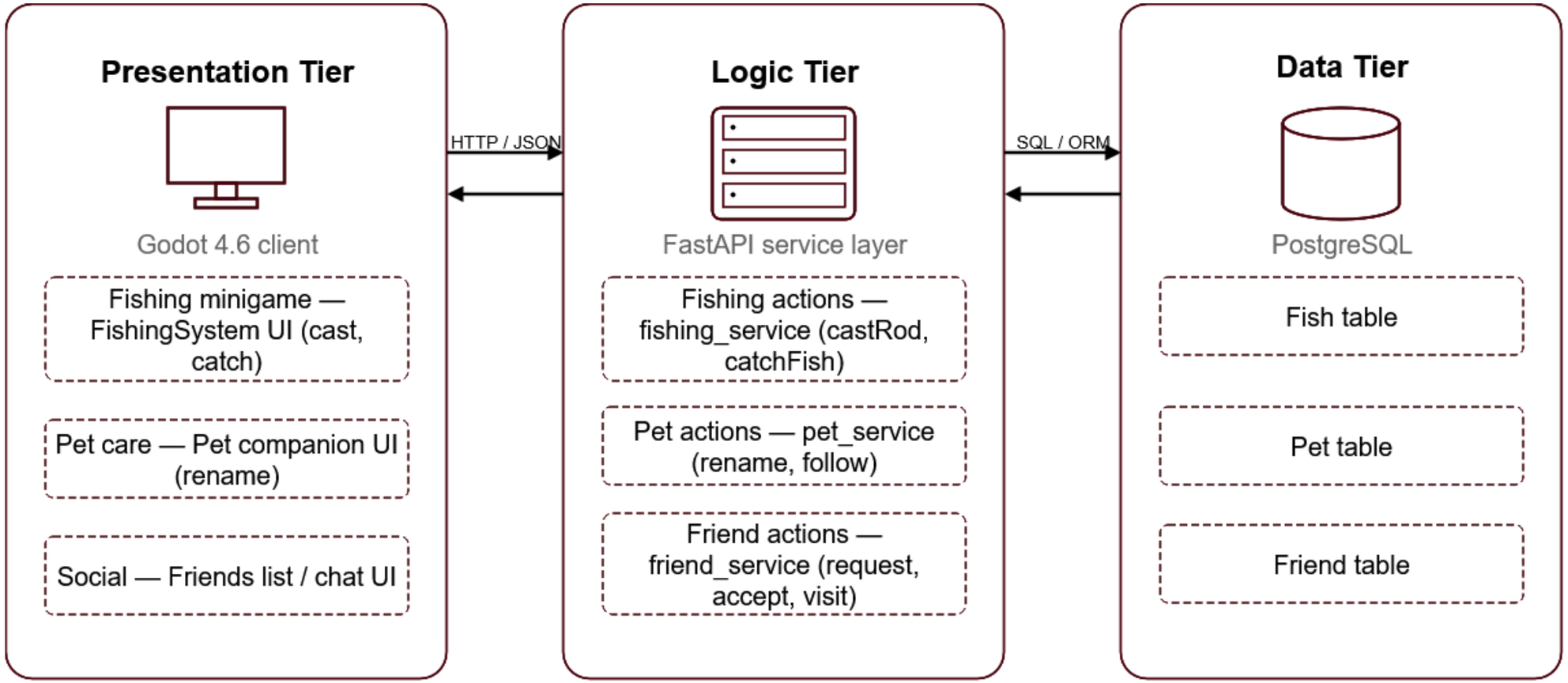
Manages the shared map, tile-based farming (till/plant/water/harvest), and the season/weather cycle that affects crop growth.

Architecture: Domain 3
Inventory, Economy & Shop



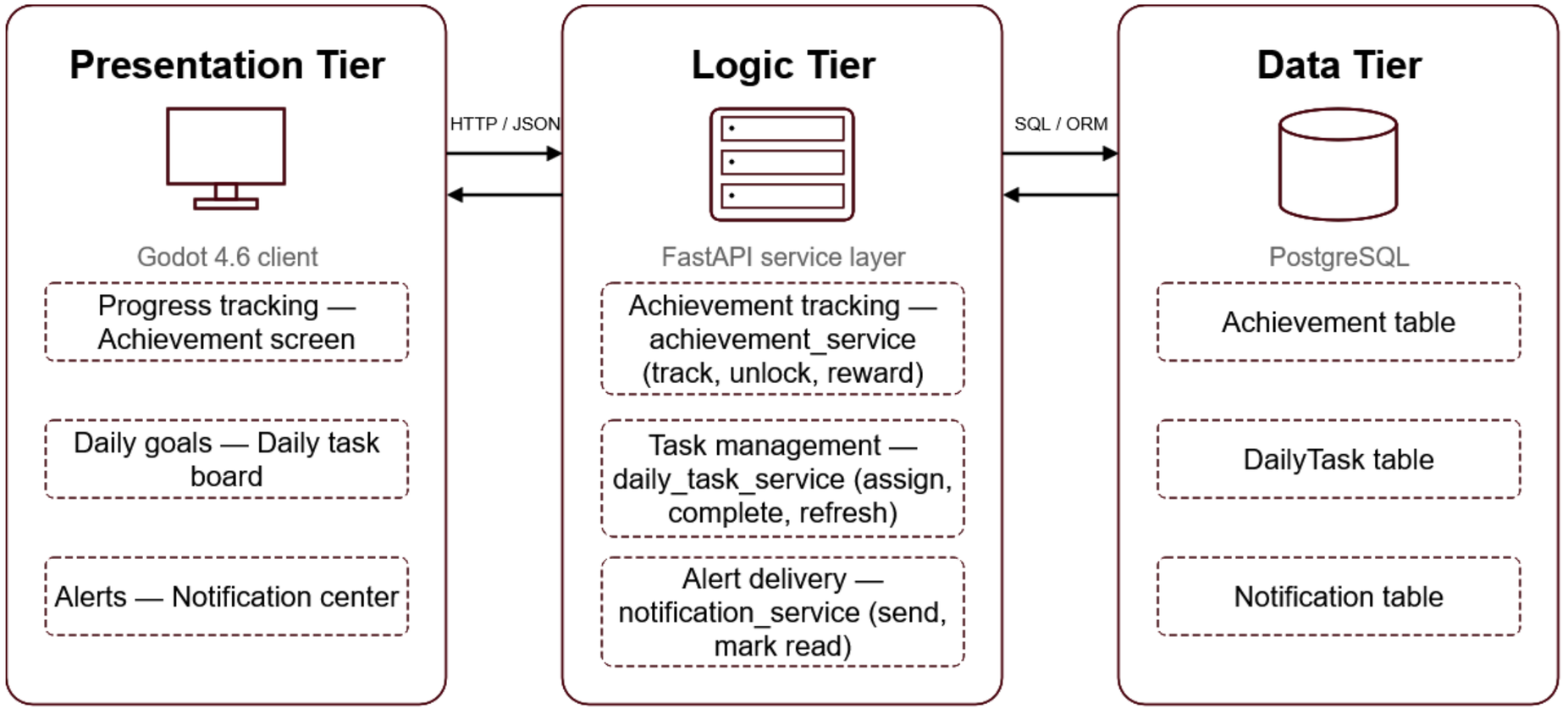
Covers item storage across partitions, the Aura currency and XP system, and shop purchases (shop still planned).

Architecture: Domain 4
Fishing, Pets & Social



Covers the fishing mini-game, pet companion, and friends/chat features — fully designed in the SRS but not yet implemented

Architecture: Domain 5
Achievements, Daily Tasks & Notifications



Covers player progression loops and in-game alerts (achievements, daily tasks, notifications) — also planned but not yet built.

THANK YOU

